

Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester II Examination April 2010

NT709 Fabrication of Nanomaterials-I

Date: 29th April 2010

Time: 10.00 am to 10.30 pm

Max. Marks 20

Part – A

Do as directed (Show your calculation within the space provided)

Que.1 when the size of semiconductor materials reaches de Broglie's wavelength the band gap : (1)
(i) Increases (ii) Decreases (iii) remains same

Que.2 The high indexed facets have: (1)
(i) Low surface energy (ii) High surface energy (iii) Independent of surface energy

Que.3 The surface energy for {110} facet will be (show the calculation and tick the correct answer) (3)

(i) $4\epsilon/a^2$ (ii) $(5/\sqrt{2}) \epsilon/a^2$ (iii) $(2\sqrt{3}) \epsilon/a^2$ (iv) ϵ/a^2

Que.4 Sintering becomes important when materials are heated typically of the melting point. (Tick the correct one) (1)

(i) Above 50% (ii) Below 50% (iii) Above 70% (iv) Below 70%

Que.5 Surface energy of convex surface is : (1)
(i) Positive (ii) Negative (iii) does not depend

Que.6 At room temperature the surface charge density is given by: (1)

(i) $E \sim 0.6[(p.z.c)-pH]$ (ii) $E \sim 0.06[pH-(p.z.c)]$ (iii) $E \sim 0.06[(p.z.c)+pH]$ (iv) $E \sim 0.06 [(p.z.c)-pH]$

Que.7 Calculate the van der Waals attraction between two blocks separated by a distance 2 nm and having the Hamaker's constant 3.14×10^{-20} J (3)

Que.8 Electrostatic stabilization is limited to: (1)

- (i) Concentrated system (ii) Multiphase system (iii) Dilute system

Que.9 If the super saturation of the system is 1.2 at 100K and the atomic volume is 1380 nm^3 the value of change in Gibbs free energy per unit volume is approximately (Show calculation and tick the correct one) (3)

- (i) 593 J/m^3 (ii) 693 J/m^3 (iii) 493 J/m^3

Que.10 The diffusion growth rate is given by $dr/dt = k_D/r$, where r is the radius of spherical nucleus. If initial radius of the nucleus is 2 nm and $k_D = 2 \times 10^{-18} \text{ /sec}$, the growth of the spherical particle after 6 sec is (Show calculation and tick the correct one) (3)

Que.11 In poly growth mechanism the growth rate is a function of (1)

- (i) Particle size (ii) Temperature (iii) Time.

Que. 12 A strong reduction promotes: (1)

- (i) Slow reaction rate (ii) fast reaction rate (iii) does not influence

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Part B

(Write each section on separate answer book)

Section I

- Que 1. (a) The random doping fluctuation becomes extremely important at nanoscale, explain the statement with example. (3)
- (b) What are the challenges one must be met in fabrication and processing of nanomaterials? (3)
- (c) Write the difference, in terms of approach, between top-down and bottom-up methods of synthesis of nanoparticles. (4)
- Que 2. (a) How to determine the shape/surface of an equilibrium crystal using Wulff plot, explain with example. (4)
- (b) Briefly explain the dependence of solubility as a function of surface curvature. (2)
- (c) By which mechanisms a surface charge will be developed when a solid merges in a polar solvent or an electrolyte solution? (4)

Section II

- Que 1. (a) What are the ways to reduce the surface energy for a given surface with fixed surface area? Discuss with example. (4)
- (b) The process of sintering and Ostwald ripening, both reduces overall surface energy, discuss under what condition sintering process takes place and when Ostwald ripening dominates. (2)
- (c) What are the basic assumptions in DLVO theory and under which conditions one can apply this theory? (4)
- Que 2. (a) Explain the process of synthesis of nano-particles inside micro emulsion. (5)
- (b) Derive the critical energy barrier expression for a heterogeneous nucleation. (5)

OR

- Que 2. (a) Draw the flow chart of synthesis of semiconductor nanoparticles using chemical technique and explain in brief. (4)
- (b) How reducing agents influence size and size distribution of metallic colloids? (3)
- (c) What are advantages of polymeric stabilization over electrostatic stabilization? (3)
- Que 3. (a) Draw the solute concentration versus time plot and explain using the plot the process of nucleation and growth. (4)
- (b) In a system, the change in Gibb's free energy per unit volume of solid phase increases by a factor of two and surface energy reduces by same factor, then calculate the change in critical energy barrier due to these changes and what should be minimum size of a stable spherical nucleus. (3)
- (c) Name the factors affecting the size distribution of nanoparticles. Which growth process results into uniform sized particles? (3)
